

Ujjwal Thapliyal

Data Scientist

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PROFILE

Data Scientist with an IIT Madras Pravartak certification and hands-on experience building end-to-end ML pipelines, NLP systems, Hypothesis Testing, and scalable PySpark workflows. Delivered production-grade models across fraud detection, customer sentiment, and satisfaction prediction — raw data to deployed API. Targeting applied DS and ML roles in BFSI, fintech, and product analytics.

PROFESSIONAL EXPERIENCE

AI ML Intern | TechnicalOne (Remote)

Jan 2026 – Feb 2026

- Built a fake news detection system end-to-end: collected and cleaned a multi-source dataset (**70K+ records across WELFake**), fine-tuned DistilBERT for binary classification achieving **82% test accuracy**, and wrapped the inference layer in a FastAPI application containerised with Docker.
- Engineered large-scale text preprocessing pipelines — tokenisation, de-duplication, stratified sampling, and label balancing — ensuring data quality at every stage before model fine-tuning.
- **Deployed** the production-grade model to **Hugging Face Spaces** as a live public API, completing full delivery from raw data to interactive web application.

PROJECTS

Credit Card Fraud Detection , Python | PySpark | XGBoost | SMOTE | SHAP

Jun 2026 – Jun 2026

- Engineered a **7-stage fully automated pipeline on the MLG-ULB dataset (284,807 real transactions, 0.172% fraud rate)** — schema validation and MLlib for scalable preprocessing (StandardScaler, stratified splits, null audits); pipeline architecture scales to 10x data volume without code changes — demonstrating big data environment experience.
- **Addressed extreme class imbalance (0.172% fraud) with SMOTE oversampling** and XGBoost class weighting; selected decision threshold via F2 score + business cost matrix (penalising missed fraud over false positives) — **AUC-ROC ~0.98, AUC-PR ~0.85**.
- **Generated SHAP explanations** per transaction (top-3 contributing features per score) to produce auditable, stakeholder-readable outputs — equivalent to risk and opportunity reporting for business decision-making.

Swiggy & CRED — Review Sentiment & Changepoint Analysis ,

Jun 2026 – Jun 2026

Product Analytics | VADER | PELT | TF-IDF | Mann-Whitney U | Python

- Detected a statistically significant **1.3-star rating decline for Swiggy using PELT changepoint detection on 25K+ Play Store reviews**, validated by Mann-Whitney U ($p < 0.05$) and Cohen's d — pinpointing rating declines correlated with specific feature rollouts.
- **Extracted top complaint keywords** per changepoint window via TF-IDF, delivering product health insights that link sentiment shifts to identifiable app changes.

FlipSatisfy – Customer Satisfaction Prediction , NLP | FastAPI | Docker

Jul 2025 – Aug 2025

- Designed an **end-to-end ML pipeline on 85,907 raw** Flipkart customer support records; applied TF-IDF vectorisation and label encoding to convert unstructured text into analysis-ready features.
- Trained and benchmarked multiple classifiers (Random Forest, AdaBoost, KNN); improved accuracy from **95.6% to 96.9%** through systematic hyperparameter tuning — every experiment tracked in MLFLOW.
- Deployed the optimised model as a containerised FastAPI microservice on Docker for real-time satisfaction inference; integrated a visual reporting layer for operational stakeholders to monitor and act on predictions.

Flight Price & Customer Satisfaction Prediction System ,

Jun 2025 – Jul 2025

Machine Learning | Python | Scikit-learn | XGBoost | MLflow | Streamlit

- Analyzed and engineered features from **110K+ flight records** to uncover pricing and booking trends.
- Developed Linear Regression, Random Forest, and XGBoost models (**$R^2 = 0.85$, 23% RMSE reduction**).
- Built passenger satisfaction models on **10K+ records (80% accuracy, 0.80 F1-score)** and performed customer segmentation.
- **Deployed a Streamlit + MLflow** solution for model serving, tracking, and stakeholder reporting.

SKILLS

Language and Core AI / ML — Python, SQL, Machine Learning, Deep Learning, Neural Networks (CNN, RNN), Statistical Modelling, Hypothesis Testing, Feature Engineering, EDA

Libraries and Framework — NumPy, Pandas, Scikit-learn, XGBoost, SciPy, Statsmodels, Matplotlib, Seaborn, Plotly, NLTK

NLP & Text — Natural Language Processing (NLP), Transformers, Hugging Face, TF-IDF, VADER, Sentiment Analysis, PELT Changepoint Detection

MLOps & Deployment — MLflow, FastAPI, Docker, Streamlit, Hugging Face Spaces, Git

Data Science — Data Analysis, Feature Engineering, EDA, Statistical Modelling, Hypothesis Testing (Mann-Whitney U, Cohen's d), A/B Testing, Customer Segmentation

Generative AI & AI Tools — LLMs, Hugging Face Inference APIs

Domain Exposure — Fraud Detection, Customer Churn, Sentiment Analysis, Travel Analytics, Revenue Optimisation, Customer Segmentation

EDUCATION

Master of Computer Applications (MCA), SGRR University Dehradun 

Jul 2024 – Present | Dehradun Uttarakhand

Master Data Science Program, IIT Madras Guvi HCL 

Nov 2024 – Jun 2025 | Remote

Bachelor's Degree 2021–2024, SGRR University Dehradun 

Jul 2021 – Jul 2024 | Dehradun Uttarakhand

CERTIFICATES

Data Science Certification Guvi HCL : IIT Madras 